

Jamal Ghobrial

AI Engineer | Machine Learning | Intelligent Systems & Automation

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Professional Summary

AI Engineer with a BSc in Bioinformatics (GPA 3.61/4.0, Very Good with Honors) and 19+ months across 4 internships building production AI systems end-to-end. Delivered 5 projects covering automated judging, legal research, large-scale document search, multimodal e-commerce discovery, and autonomous content creation – each going from raw data to deployed system. Strong collaborator in agile, cloud-based teams; competitive machine learning practitioner (Top 29% Kaggle globally). Certified by NVIDIA, Microsoft, and WorldQuant University.

Technical Skills

Languages & Libraries: Python (PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, SciPy), SQL, Go, C++

ML & Ranking: LightGBM LambdaRank, XGBoost, Feature Engineering, Learning-to-Rank (LTR), NDCG Optimisation, ML Pipelines

LLM & Agentic AI: LangGraph StateGraph, Groq SDK, OpenAI API, Anthropic Claude API, Gemini API, OpenRouter, Prompt Engineering

RAG & Vector Search: RAG Pipelines, ChromaDB, Qdrant, pgvector, Hybrid Retrieval (BM25 + Dense RRF), SentenceTransformers (MiniLM-L6-v2, BGE-M3), Tantivy BM25

Computer Vision & Speech: YOLOv11 Pose, ByteTrack, DINOv2 (ViT-S/14), Whisper STT (Groq / Faster-Whisper), PyMuPDF, python-pptx, MoviePy

Web & Full-Stack: FastAPI, Express.js, React 19, Node.js, MongoDB / Mongoose, JWT Auth, Multer, Vite

Tools & Platforms: LibreChat, Tavily Search, Firecrawl, Jina AI Reranker, SymSpell, Mixture-of-Agents (MoA), FPDF2, Docker Compose

Cloud & DevOps: Microsoft Azure, AWS SageMaker, Docker, Modal Labs (A100 / T4 GPU), Git / GitHub

Experience

AI Engineer Intern – Areeb Technology, Cairo EG

Feb 2026 – May 2026 (4 months)

- Built OmniFind, a multimodal e-commerce search platform where users find products by image, voice, or text query across 9,842 items – using DINOv2 visual embeddings, BGE-M3 hybrid RRF semantic search, and Groq Whisper STT on a React 19 + Express + MongoDB + Qdrant stack with JWT role-based access control.
- Developed Sematrix, a 2-stage search engine indexing 8.8M MS MARCO documents – Tantivy BM25 recall with SymSpell spell-correction, re-ranked by LightGBM LambdaRank (7 engineered features) on Modal A100 GPU – achieving NDCG@10 improvement over BM25 baseline.
- Architected ContentPro Creator Agent, a 3-model Mixture-of-Agents pipeline on LibreChat: Gemini 2.5 Pro (research + Tavily/Firecrawl/Jina) to Claude Sonnet 4 (writing + LibreChat Artifacts) to GPT-4.1 (QA editor), with pgvector RAG for brand-voice consistency.

AI Engineer Intern – NTL, Cairo EG

Oct 2025 – Jan 2026 (4 months)

- Designed Digitopia AI Competition Judgment System: Phase 1 extracts audio (MoviePy + Groq Whisper STT) and tracks body language per presenter (YOLOv11 Pose + ByteTrack, scoring gesture rate, posture variance, and head movement); Phase 2 runs a LangGraph StateGraph grading agent (Plan, Score, FactCheck via DuckDuckGo, Verify with self-correction) over PyMuPDF + python-pptx document inputs.
- Integrated ChromaDB RAG (MiniLM-L6-v2) for natural-language jury Q&A over all evaluation data, SQLite persistence, and FPDF2 automated PDF report generation; benchmarked ML, CV, and NLP models across PyTorch and TensorFlow.

ML Engineer Intern – Microsoft Track – DEPI, Cairo EG

Jun 2025 – Dec 2025 (7 months)

- Trained classification, regression, and clustering models (Scikit-Learn, TensorFlow) following MLOps best practices; automated ETL pipelines to reduce data preparation time across iterative experiment cycles.

Machine Learning Intern – ITIDA / EME Initiative, Assiut EG

Jul 2023 – Oct 2023 (4 months)

- Built an industrial defect prediction model (XGBoost + Scikit-Learn) achieving **85% accuracy** with a **40% performance uplift** via dimensionality reduction and hyperparameter tuning; placed **Top 29%** and **Top 33%** in global Kaggle ML competitions.

Featured Projects

Digitopia AI Competition Judgment System 🌐

LangGraph, YOLOv11, Groq, ChromaDB

- AI judging system that automatically scores hackathon teams by analysing pitch videos (Groq Whisper STT + YOLOv11 Pose body-language) and project documents (LangGraph self-verifying grading agent: Plan, Score, FactCheck, Verify), with a ChromaDB RAG jury assistant and FPDF2 PDF report export.

LexAI: Legal Document Assistant 🌐

FastAPI, MiniLM-L6, BM25, OpenRouter, SSE

- Legal research assistant with Hybrid Retrieval (BM25 + MiniLM-L6-v2 dense + RRF), section-aware chunking at Article/Clause boundaries, real-time SSE token streaming, source citations, cross-document querying, and DOCX

export – no LangChain or FAISS dependencies.

- Semantrix: Learning-to-Rank Search Engine***Tantivy, LightGBM LambdaRank, SentenceTransformers, Modal*
- 2-stage LTR pipeline over 8.8M MS MARCO documents: Tantivy BM25 + SymSpell recall, re-ranked by LightGBM LambdaRank with 7 features (BM25 proxy, TF-IDF proxy, overlap ratio, exact match, semantic cosine via MiniLM-L6-v2), deployed serverlessly on Modal A100/T4 GPU.
- OmniFind: Multi-Modal Product Discovery Platform***DINOv2, BGE-M3, Qdrant, Express.js, React 19*
- Full-stack e-commerce search supporting image (DINOv2 384-dim cosine), text (BGE-M3 1024-dim dense+sparse RRF), and voice (Groq Whisper) search over 9,842 products, with Cloudflare Workers AI intent parsing, score boosting, JWT auth, and Docker + MongoDB 7 stack.
- ContentPro Creator Agent***LibreChat, Gemini 2.5 Pro, Claude Sonnet 4, GPT-4.1*
- 3-model Mixture-of-Agents content pipeline: Gemini 2.5 Pro (web research via Tavily + Firecrawl + Jina reranker) to Claude Sonnet 4 (platform-adapted writing, LibreChat Artifacts) to GPT-4.1 (QA: CTA, hashtag, word-count checks), with pgvector RAG for brand guidelines.

Education

BSc in Bioinformatics – GPA **3.61 / 4.0** (Very Good with Honors)**Jul 2023**

Faculty of Computers & Information, Assiut University, Egypt – Graduation Project recognised for excellence.

Courses & Certifications

Getting Started with Deep Learning – NVIDIA Deep Learning InstituteDec 2025

Applied Data Science Lab – WorldQuant UniversityJul 2023

Data Analysis Professional Nanodegree – UdacityDec 2022

Additional Information

Languages: Arabic (Native) English (Professional Working) **Military Service:** Completed (Jun 2025)